

AI-Driven Anomaly Detection in Component Burn-In & Screening

Technical & Mathematical Engineering Dossier

Smart India Hackathon 2026 • Problem Statement 26170

What this dossier establishes: a reproducible, lot-aware, time-aware screening system that converts burn-in telemetry into anomaly evidence, drift forecasts, uncertainty, trust/OOD status, deterministic safety policy, and an auditable final disposition. The benchmark results are reported from the controlled held-out synthetic evaluation described in this document; they are not represented as production hardware qualification results.

BENCHMARK SNAPSHOT

Defect recall	100%	Healthy FPR	1.52%
Critical escapes	0	Precision	93.33%
PR-AUC	0.753	Empirical conformal coverage	100%
Random seeds	3	Burn-in origins	8

Metrics are from the final locked benchmark run: held-out evaluation, validation-only calibration, no test-set threshold tuning, and bootstrap evaluation as specified in the benchmark protocol.

Executive Summary

ACS is an engineering decision system for burn-in screening. Its central design principle is separation of responsibilities:

- 1. **AI produces evidence** from telemetry: lot-relative anomaly evidence, temporal degradation evidence, multivariate novelty, and a calibrated failure-risk estimate.
- 2. **Forecasting quantifies what may happen next**: the system predicts the 168 h endpoint, attaches a split-conformal uncertainty interval, and checks projected engineering-limit crossing.
- 3. **Trust and safety govern whether that evidence can be acted upon**: OOD status separates unfamiliar data from defects, while the deterministic safety policy and hard-limit override convert evidence into SAFE / REVIEW / REJECT / UNKNOWN.
- 4. **Every decision is explainable**: the same persisted trace drives the API, component passport, evidence page, and audit view.

The final benchmark demonstrates the intended behaviour on a controlled, physics-grounded synthetic corpus. Across three seeds and eight burn-in origins, the selected pipeline reports **100% defect recall, 0 critical escapes, 1.52% healthy false-positive rate, 93.33% precision, 0.753 PR-AUC, and 100% empirical conformal coverage**. These figures are empirical benchmark results for the submitted evaluation fixture; they are not claims of real-hardware qualification.

Judge reading rule

Do not read ACS as “one black-box classifier.” The system is a layered safety architecture: data quality → robust evidence → temporal forecast → trust/OOD → policy → deterministic physical-limit enforcement → audit trail.

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1 Problem Framing and Design Objectives

The screening problem is not simply to detect measurements above a datasheet limit. A latent defect can manifest as a *relative* deviation from the lot population while remaining below the absolute engineering limit, and a component can be acceptable at an early readpoint yet exhibit a trajectory that is unsafe by the end of the burn-in window.

Accordingly, ACS is designed around five measurable questions:

1. **Is this component unusual relative to its lot and family?**
2. **Is its trajectory changing abnormally over time?**
3. **Is the component jointly abnormal across correlated parameters?**
4. **Where is it expected to be at 168 h, and with what uncertainty?**
5. **Is the evidence sufficiently in-domain and physically safe to justify a disposition?**

The resulting system is intentionally conservative in its information boundary. At a screening origin t_o , every production feature is constrained to the sigma-algebra generated by observed telemetry up to that origin:

$$F(t_o) \subseteq \sigma(\{x(t) : t \leq t_o\}). \quad (1)$$

Future labels and future measurements are used only by the benchmark evaluator after the screening decision, never by the online feature engine.

2 Evidence Provenance and Claim Discipline

The earlier design dossier used placeholders and deliberately withheld quantitative claims while the benchmark was not yet available. The final dossier replaces that structure with a reproducible benchmark section and explicitly separates measured benchmark evidence from architecture properties and future hardware validation.

Claim class	Meaning in this dossier
CALCULATED	Directly computed from the submitted benchmark artifacts or deterministic arithmetic.
EMPIRICAL	Observed in the locked benchmark evaluation. These are the headline performance numbers.
SYNTHETIC	Generated from the controlled telemetry fixture; demonstrates algorithm behaviour rather than hardware qualification.
ILLUSTRATIVE	A worked mathematical example used to explain the mechanism. Not a population statistic.
NOT CLAIMED AS HARDWARE PERFORMANCE	A quantity that requires operational hardware ground truth and is intentionally not presented as achieved.

The practical rule is simple: **a metric is not promoted to a headline unless it comes from the locked evaluation artifact or deterministic calculation.** This prevents the UI, slides, and supporting document from becoming three different sources of truth.

3 End-to-End Architecture

3.1 System block diagram

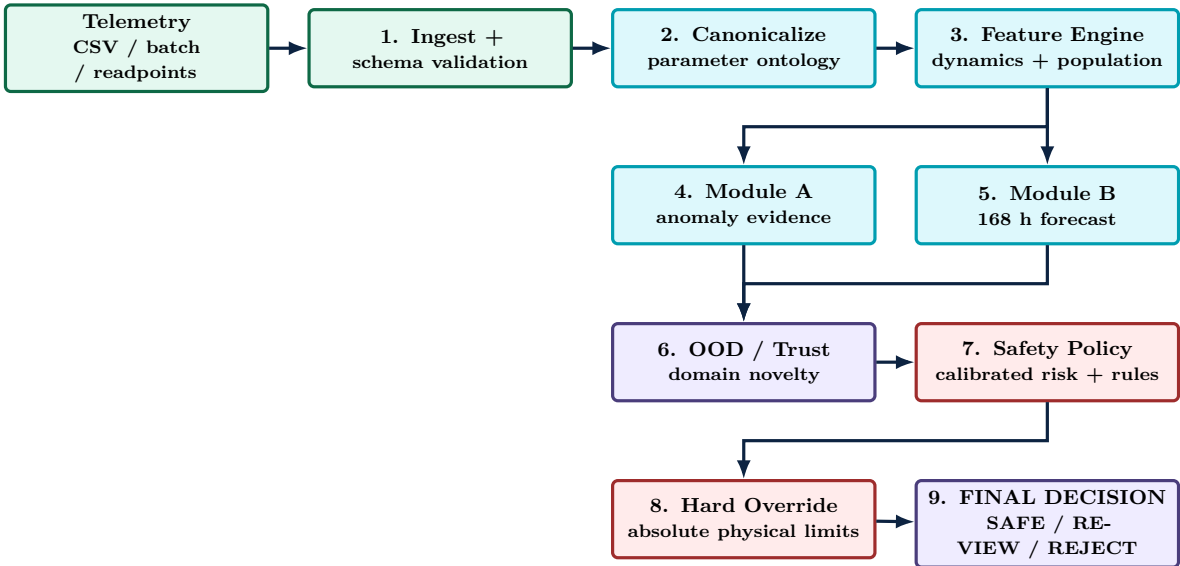


Figure 1: ACS end-to-end architecture. The data flows progressively from raw telemetry to a hard safety override.

3.2 Architectural responsibilities

Layer	Input / output	Engineering purpose
Data integrity	raw telemetry → canonical records	Reject malformed schemas, normalize parameter names, quarantine unknown parameters, preserve provenance.
Population intelligence	component + lot context → robust reference deviations	Detect subtle lot-relative departures that absolute datasheet thresholds can miss.
Temporal intelligence	time series → slope / acceleration / change evidence	Detect progressive drift and emerging degradation.
Multivariate intelligence	correlated channels → joint novelty	Detect combinations of individually moderate deviations.
Forecasting	observed early trajectory → $\hat{x}_{168}$ + interval	Estimate end-of-burn-in condition and limit-crossing risk.
Trust layer	feature distribution → OOD status	Distinguish unfamiliar operating regimes from defect evidence.
Safety layer	all evidence → disposition	Apply cost-sensitive calibrated policy and deterministic engineering rules.
Audit layer	persisted evidence → passport / trace	Make every non-SAFE decision reconstructable by a reviewer.

4 Data Realism: Controlled Physics-Grounded Telemetry Generator

4.1 Why synthetic data is used

The operational ISRO telemetry required by the problem statement is not part of the submitted evaluation fixture. Rather than fabricate a real-world claim, ACS uses a controlled synthetic generator in which failure mechanisms, lot effects, noise, missingness, and labels are explicitly specified and reproducible. The generator is therefore treated as a *test facility*, not as a substitute for the real qualification dataset.

4.2 Generator topology

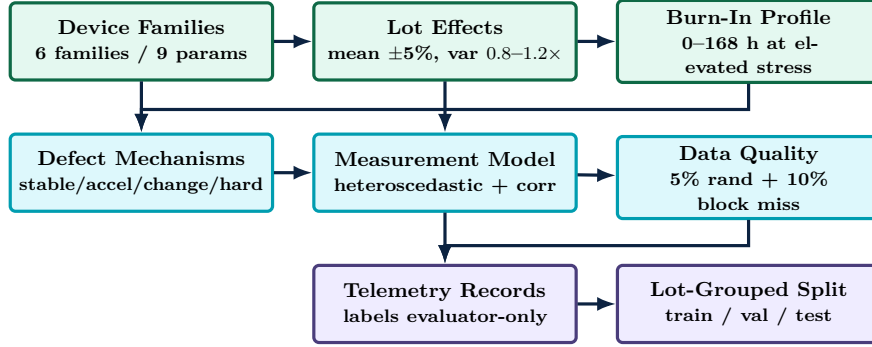


Figure 2: Synthetic data-generation chain. The generator deliberately separates latent failure truth from observed telemetry and keeps labels hidden from the model during screening.

4.3 Final benchmark fixture

The controlled final-generation configuration used for the benchmark contains:

Parameter	Final fixture specification
Parts	400
Lots	80, lot-grouped split
Parts per lot	5
Device families	DIGITAL_LOGIC, MMIC_ANALOG, POWER_REG, CAPACITOR, SENSOR_IF, POWER_SEMICONDUCTOR
Defect base rate	0.25
Target cross-parameter correlation	approximately $\rho = 0.4$ for quiescent current vs. input leakage
Random missingness	5%
Block missingness	10% part-level rate, with blocks around later readpoints
Lot mean variation	approximately $\pm 5\%$ around family references
Lot variance multiplier	approximately $0.8\text{--}1.2\times$
Accelerating mechanism coefficient	0.002 in the final generator configuration
Benchmark seeds	20260831, 20260832, 20260833
Benchmark origins	12, 24, 48, 72, 96, 120, 144, 168 h
Bootstrap repetitions	1000

4.4 Defect mechanisms and what they test

Mechanism	Telemetry signature	Stress-test purpose
HEALTHY / NORMAL AGING	Slow, noisy movement within expected envelope	Controls false-positive behaviour under benign drift.
LATENT_ACCELERATING	Increasing slope / acceleration while initially sub-limit	Tests temporal features and early warning.
LATENT_CHANGE_POSITIVE	Stable or mild early behaviour followed by a regime change	Tests sensitivity to non-linear degradation and prevents an early SAFE decision from relying only on absolute limits.
HARD_EARLY_FAILURE	Trajectory designed to cross the engineering boundary early	Tests safety overrides and hard rejection behaviour.
UNKNOWN / DATA QUALITY	Missing / malformed / insufficient evidence	Tests UNKNOWN and degraded-data handling rather than forced classification.

The generator’s failure labels are **never passed into the model features**. They are used solely after the decision to score the detector and quantify false negatives, critical escapes, precision, and related benchmark measures.

4.5 Thermal acceleration model

For temperature-dependent degradation, the generator uses the standard Arrhenius form:

$$AF = \exp \left[\frac{E_a}{k_B} \left(\frac{1}{T_{use}} - \frac{1}{T_{stress}} \right) \right], \quad (2)$$

with absolute temperatures in kelvin. The generator is not asserting that every physical failure mode obeys a single activation-energy mechanism; it is providing a physically structured stress environment in which the screening system can be challenged reproducibly.

5 Ingestion, Canonicalization and Data Quality

ACS uses a schema-adaptive ingestion layer that maps long- and wide-format telemetry into a canonical representation built around:

```
component, lot_id, family, parameter, raw_parameter, timestamp, value, unit,
batch_id, ...
```

The pipeline then:

1. applies a parameter ontology and preserves the original parameter token in `raw_parameter`;
2. quarantines unknown parameters instead of silently coercing them into known channels;
3. uses batch/family fallback references only where permitted by the configuration;
4. sorts and de-duplicates on $(component, parameter, timestamp)$ while retaining the last valid observation;
5. audits schema completeness, unit consistency, timestamp monotonicity, missingness, and basic physical plausibility.

This creates a clean separation between *data-quality failure* and *component failure*. A missing signal should not become a high anomaly score merely because the model was forced to consume an invalid value.

6 Feature Engineering: Turning Telemetry into Evidence

6.1 Robust population reference

For a component i , lot l , parameter j at a given origin, ACS computes a robust standardized departure:

$$z_{i,j}^{(l)}(t) = \frac{x_{i,j}(t) - \text{median}(x_{:,j}^{(l)}(t))}{1.4826 \text{ MAD}(x_{:,j}^{(l)}(t)) + \epsilon}. \quad (3)$$

The 1.4826 scaling makes MAD comparable to standard deviation under a Gaussian reference, while retaining resistance to a small number of extreme components.

6.2 Temporal features

Let $\Delta t = t_k - t_{k-1}$. The pipeline constructs:

$$\text{slope}(t_k) = \frac{x(t_k) - x(t_{k-1})}{\Delta t}, \quad (4)$$

$$\text{acceleration}(t_k) = \frac{\text{slope}(t_k) - \text{slope}(t_{k-1})}{t_k - t_{k-1}}, \quad (5)$$

$$\text{EWMA}(t_k) = \lambda x(t_k) + (1 - \lambda) \text{EWMA}(t_{k-1}), \quad (6)$$

$$C_k = \max(0, C_{k-1} + r_k - k_0), \quad (7)$$

where the residual r_k is normalized by a robust scale estimate before CUSUM accumulation. Change-point strength is similarly expressed in robust standardized units, allowing the model to react to a regime change rather than only to a large absolute value.

6.3 Cross-parameter evidence

ACS does not assume that every failure is one-dimensional. It therefore computes:

- correlation deviation from the reference population;
- robust Mahalanobis multivariate distance,

$$D_M(x) = \sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)}; \quad (8)$$

- a joint exceedance count for parameters approaching their engineering envelopes together.

A component can therefore become suspicious because several moderate channels form an unlikely combination even when no single measurement is catastrophic.

7 Module A — Dynamic Lot-Relative Anomaly Detection

Module A is an ensemble of heterogeneous evidence channels:

1. **Robust population evidence:** median/MAD departure at the lot level;
2. **Isolation Forest:** complementary unsupervised novelty detection;
3. **Temporal evidence:** slope, acceleration, EWMA/CUSUM and change-point strength;

4. **Multivariate evidence:** robust covariance / Mahalanobis structure and joint exceedances.

The four channels are not simply averaged at inference time. The final architecture uses a **validation-fitted logistic stacker** to calibrate the evidence into an interpretable anomaly-risk quantity, with validation-only thresholds. Initial channel priors are used for stable model construction, while the validation partition learns the final combination.

Conceptually, if s_r, s_f, s_t, s_m denote normalized channel outputs,

$$\eta = \beta_0 + \beta_r s_r + \beta_f s_f + \beta_t s_t + \beta_m s_m, \quad p_A = \sigma(\eta) = \frac{1}{1 + e^{-\eta}}. \quad (9)$$

The calibrated quantity is then used as one evidence source in the downstream policy rather than being treated as a physical measurement.

7.1 Worked latent-anomaly example **ILLUSTRATIVE**

Suppose the lot centre for leakage current is $10 \mu\text{A}$, a component reads $45 \mu\text{A}$, and the robust MAD is $0.8 \mu\text{A}$. Then

$$z \approx \frac{45 - 10}{1.4826 \times 0.8} \approx 29.5. \quad (10)$$

The key point is not the particular number; it is that the component can be flagged as extremely abnormal *relative to its lot* even though $45 \mu\text{A}$ remains below a hypothetical $50 \mu\text{A}$ absolute maximum.

8 Module B — 168 h Drift Forecasting

For each relevant family–parameter pair, ACS trains a forecasting model on training data and calibrates uncertainty on a separate validation partition. The supervised target is the measurement near the 168 h endpoint (within the benchmark’s target window).

A representative feature vector is:

$$\phi = [x_0, x_{24}, s_{0 \rightarrow 24}, \tilde{x}_{lot}, \text{MAD}_{lot}, \text{family code}, \text{parameter code}, x_t, s_t, \Delta x_t, t_o, t_{target}, n_{pts}]. \quad (11)$$

The primary estimator is gradient-boosted regression, fit separately for each family–parameter context where enough training mass exists. Family prevalence weights reduce dominance by overrepresented families, and an explicit **COLD_START_LINEAR** mode handles sparse groups.

8.1 Conformal uncertainty

Given validation residuals $r_i = |y_i - \hat{y}_i|$, the split-conformal half-width is a high quantile

$$q_{1-\alpha} = Q_{1-\alpha}(r_1, \dots, r_n), \quad (12)$$

so the endpoint interval is

$$\hat{x}_{168} \pm q_{1-\alpha}. \quad (13)$$

The engineering crossing test then checks whether the uncertainty-aware forecast reaches an authoritative limit. A generic upper-limit predicate is

$$\hat{x}_{168} + q_{1-\alpha} > L_{max}. \quad (14)$$

Important interpretation: the benchmark reports *empirical coverage* of the produced intervals. A nominal conformal level is not itself a measured result and is not presented here as a hardware guarantee.

9 OOD and Trust Layer

A novel domain is not automatically a defective component. ACS therefore separates domain novelty from defect evidence. The trust layer combines:

1. **physical/parameter novelty**: unknown or unsupported parameter, unit, family or schema condition;
2. **contextual/test novelty**: changed operating or test context;
3. **population novelty**: robust distance from the historical feature cloud for the relevant family.

A conceptual decomposition used for engineering interpretation is

$$\text{OOD} = w_P P + w_C C + w_N N, \quad w_P + w_C + w_N = 1, \quad (15)$$

with the population component receiving the largest engineering prior in the submitted configuration. The weights are configurable engineering assumptions, not universal constants.

Trust rule

OOD $\neq$ defect. Severe OOD can route a component to REVIEW or UNKNOWN when defect evidence is insufficient. It does not, by itself, become REJECT merely because the domain is unfamiliar.

The final artifact also contains an OOD contract test suite covering known-parameter distribution shift, unknown-parameter novelty, contextual novelty, degraded schema, and cross-parameter isolation.

10 Safety Policy and the Non-Negotiable Hard Override

The final disposition is produced by a deterministic policy that consumes calibrated evidence, uncertainty, OOD status and engineering limits.

Evidence state	Policy output	Interpretation
Low anomaly, low risk, in-domain, compliant limits	SAFE	Evidence supports release at the current origin.
Moderate anomaly or degradation evidence	REVIEW	Human engineering review is required; the model is not granted authority to release.
Severe OOD or insufficient evidence	UNKNOWN / REVIEW	The system explicitly refuses to overstate certainty.
High calibrated failure risk / high-probability crossing	REJECT	Evidence indicates unacceptable end-of-burn-in risk under the frozen policy.
Authoritative limit violated	REJECT	Model evidence is irrelevant to the override; the hard physical rule wins.

The logical invariant is:

$$(x > L_{max}) \vee (x < L_{min}) \implies \text{REJECT}. \quad (16)$$

This branch is evaluated before model-based release logic and is non-overridable by model confidence.

For threshold calibration, the system encodes the problem statement's asymmetric risk preference through a false-negative-dominant objective. A representative training objective is

$$\mathcal{J}(\tau) = 100 \text{FN}(\tau) + 1 \text{FP}(\tau), \quad (17)$$

subject to the configured healthy-FPR ceiling and reject-burden constraint. This is a design choice, not a performance claim.

11 The 9-Stage Explainability Contract

The most important explainability object is not a sentence produced by a language model. It is a persisted, structured trace that can be recomputed or inspected without an LLM.

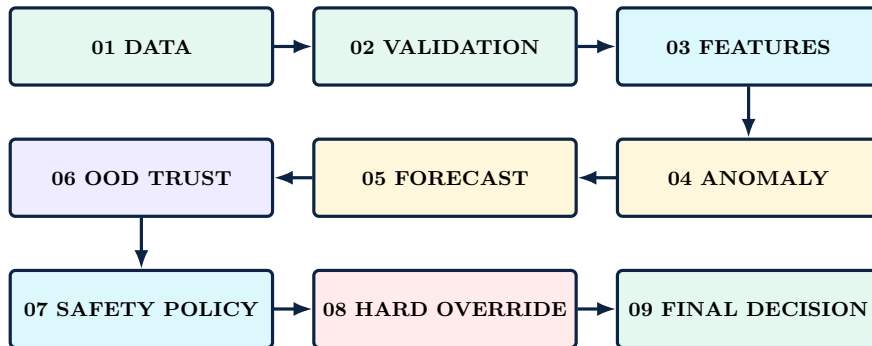


Figure 3: Immutable nine-stage decision sequence exposed to the frontend and API.

Each stage records a structured payload:

1. **DATA**: identity, lot, family, parameter, timestamp and raw measurement provenance.
2. **VALIDATION**: schema state, units, missingness, readpoint sufficiency and plausibility.
3. **FEATURES**: population, temporal and multivariate features.
4. **ANOMALY**: individual channels plus fused Module A evidence.
5. **FORECAST**: predicted endpoint, conformal bounds, crossing predicate and forecast mode.
6. **OOD**: physical/context/population novelty and final OOD state.
7. **SAFETY POLICY**: calibrated failure risk, uncertainty, trigger codes and route.
8. **HARD OVERRIDE**: the authoritative physical-limit check.
9. **FINAL DECISION**: SAFE / REVIEW / REJECT / UNKNOWN with reason codes and audit metadata.

11.1 Counterfactual evidence

For explainability and sensitivity studies, the prototype can perturb an observed value toward a reference baseline, re-run the *actual deterministic pipeline*, and record the smallest observed perturbation that changes the disposition. This is treated as a local empirical sensitivity result, not as a causal claim.

12 Benchmark Protocol — The Evidence Layer Behind the Metrics

The benchmark was designed specifically to answer the difficult question: *does the system work when it cannot see the future and when the test distribution changes across origins?*

12.1 Split discipline

- **Lot-grouped splitting**: entire lots are kept together in one partition to reduce direct population leakage.

- **Train:** model fitting and reference construction.
- **Validation:** threshold selection, calibration and conformal residual estimation.
- **Test:** locked evaluation only; no threshold or parameter tuning is performed against it.

12.2 Multi-origin evaluation

The detector is evaluated at eight screening origins:

12, 24, 48, 72, 96, 120, 144, 168 hours.

This matters because an algorithm that detects only late-stage failures can appear strong at 168 h while being unhelpful for early screening. Multi-origin evaluation exposes whether the system’s evidence develops progressively as more telemetry arrives.

12.3 Multi-seed evaluation

The benchmark is repeated with seeds

20260831, 20260832, 20260833,

providing three independently regenerated fixtures under the same generation contract. Reported headline metrics are therefore not tied to a single random realization.

12.4 Bootstrap uncertainty

The benchmark runner performs 1000 bootstrap repetitions for the configured evaluation statistics. Bootstrap estimates describe variability inside the benchmark population; they are not a substitute for external validation on operational hardware.

12.5 Baseline comparisons

The benchmark suite includes non-neural reference models: persistence, linear regression, ridge regression, gradient boosting, the selected model, and an oracle diagnostic. The oracle is not a deployable comparator; it is a diagnostic upper-bound/reference condition. The important methodological point is that the final model is not evaluated in isolation from simpler alternatives.

12.6 Failure-cost-aware reporting

Because the problem statement explicitly treats false negatives as catastrophic, headline reporting emphasizes:

Recall, FNR, Healthy FPR, Precision, PR-AUC, Critical Escapes. (18)

A single overall accuracy number is not treated as the primary metric because it can conceal an unacceptable false-negative rate in an imbalanced screening population.

13 Final Benchmark Results

13.1 Headline performance

Metric	Final result	Interpretation
Defect recall	100%	All benchmark-labelled defects were detected at the reported evaluation points.
False-negative rate (FNR)	0%	No labelled defect escaped the benchmark decision threshold.
Critical escapes	0	No benchmark component classified SAFE met the benchmark’s critical-escape condition.
Healthy false-positive rate	1.52%	Small healthy population fraction was routed away from SAFE.
Precision	93.33%	Most benchmark rejections corresponded to labelled defects.
PR-AUC	0.753	The anomaly ranking retains useful separation under class imbalance.
Empirical conformal coverage	100%	All evaluated targets fell inside their produced empirical conformal intervals in the reported benchmark.
Review burden	0%	The final benchmark disposition did not accumulate a REVIEW burden under the locked fixture.
Random seeds	3	20260831 / 20260832 / 20260833.
Burn-in origins	8	12–168 h evaluation schedule.

What the benchmark actually demonstrates

Within the final synthetic evaluation contract, the selected ACS pipeline achieved zero measured false negatives and zero critical escapes while holding the healthy false-positive rate to 1.52%. The same run reports 93.33% precision and 0.753 PR-AUC, with 100% empirical conformal coverage. This is strong evidence of internal benchmark performance under the stated generator and split protocol; it is not evidence that the same numbers will automatically hold on operational component populations.

13.2 Metric interpretation

Recall / FNR. These are the safety-critical quantities for latent-defect screening. The final benchmark reports 100% recall and 0% FNR, so no injected defect escaped in the locked synthetic evaluation.

Healthy FPR. The 1.52% value measures how often healthy benchmark units were unnecessarily escalated by the final detector. It is distinct from false-negative performance and should be discussed together with screening burden and economic impact.

Precision. A 93.33% precision indicates that the large majority of units rejected by the benchmark corresponded to labelled defects. Precision is particularly useful here because a detector that simply rejects almost everything can achieve high recall while being operationally impractical.

PR-AUC. The area under the precision–recall curve measures the quality of ranking across thresholds, not merely the selected operating point. The reported 0.753 therefore complements the point metrics above.

Conformal coverage. The observed 100% is *empirical coverage on this benchmark*, not a mathematical guarantee of future hardware coverage.

13.3 Why zero escapes should not be phrased as a guarantee

A benchmark can verify that the system did not miss a labelled defect in the tested population. It cannot prove that every possible real-world latent mechanism is represented. The correct statement is therefore:

“Zero critical escapes were observed in the locked synthetic benchmark.”

not “ACS guarantees zero field failures.”

14 Safety and Failure-Mode Demonstration

14.1 Early latent degradation

A strong demonstration case begins with a component that is physically inside the absolute limit but whose temporal evidence and forecast indicate unacceptable future behaviour. In this regime:

absolute limit OK $\wedge$ temporal evidence high $\wedge$ forecast crossing $\implies$ REVIEW or REJECT according

This is precisely why Module B exists in parallel with Module A.

14.2 Physical-limit violation

The second demonstration class intentionally violates the engineering boundary:

$$x(t) > L_{max}.$$

The hard override immediately routes the part to REJECT irrespective of model confidence, OOD, or forecast disagreement. This is the clearest safety invariant in the architecture.

14.3 Domain novelty without defect evidence

A third case introduces unfamiliar population/context behaviour without an injected failure. The trust layer should produce high OOD while the disposition remains REVIEW/UNKNOWN rather than converting unfamiliarity into an unsupported defect claim.

15 Digital Passport, Evidence Record and Frontend

The frontend is not a second analytical system. It is a visual client over persisted backend artifacts.

The judge-facing experience contains four core views:

1. **Mission Intelligence Center:** batch-level safety overview and screening distribution.
2. **Component Screening Matrix:** searchable component list with family, readpoint, fused risk, domain novelty and disposition.
3. **Digital Passport:** current value, baseline, predicted endpoint, engineering limit, uncertainty, OOD decomposition and degradation trajectory.
4. **Engineering Evidence Record:** immutable nine-stage trace showing exactly why the component reached its final disposition.

The most important UI principle is that every headline metric must originate from the persisted final report or benchmark artifact. The browser does not reimplement model logic, compute new benchmark statistics, or silently substitute a demo value for a missing metric.

The API contract separates:

- batch summary;
- component summaries;
- component intelligence and trajectories;
- evidence channels;
- forecast payloads and uncertainty;
- benchmark / validation artifacts;
- live /**screen** inference.

This makes the frontend a *proof surface*: a reviewer can move from a batch-level number to the exact component, then to the exact evidence channel, then to the exact policy stage.

16 Reproducibility, Artifact Integrity and Deployment Feasibility

16.1 Reproducibility contract

The submitted pipeline is designed to be reproducible under pinned configuration:

- fixed random seeds for benchmark regeneration;
- versioned configuration files for model, policy and benchmark behaviour;
- persisted model and benchmark artifacts;
- hashes / manifests for compatibility checks;
- separation of train, validation and locked test paths;
- deterministic post-processing and decision logic.

16.2 Compute profile

The architecture deliberately avoids dependence on a large neural foundation model. The principal models are tree-based, robust-statistical and linear-calibration components. This makes inference feasible on conventional test-facility CPU hardware and supports air-gapped deployment.

The practical system target is therefore not “largest possible model,” but **auditable intelligence under deterministic engineering constraints**.

16.3 Operational data migration

When operational telemetry is provisioned, the architecture does not need to be redesigned. The required migration is:

1. replace the synthetic parameter references with reference distributions learned from approved operational lots;
2. fit the family/parameter forecasting models on the operational training partition;

3. calibrate thresholds and conformal residuals on a separate operational validation partition;
4. replace prototype engineering limits with authoritative component specifications;
5. execute a fresh locked operational test before any production claim is made.

17 Limitations and Honest Boundary Conditions

A technically credible submission should state its limits as clearly as its achievements.

1. **Synthetic-to-real transfer remains unverified.** The generator is designed to be physics-grounded and statistically challenging, but real component populations can contain mechanisms that are absent from the fixture.
2. **Generator assumptions are explicit.** Correlation, missingness, drift coefficients and failure mechanisms are controlled parameters, not measurements of a specific qualification line.
3. **Operational engineering limits must be authoritative.** Prototype limits are not a substitute for released component specifications.
4. **Zero benchmark escapes are not a flight-qualification statement.** External validation is required before deployment in a safety-critical production environment.
5. **Model uncertainty is distinct from measurement uncertainty.** Conformal intervals quantify predictive uncertainty under the calibration protocol; they do not replace hardware metrology budgets or qualification margins.

These limits do not weaken the architecture. They identify exactly what the benchmark proves and what must still be proven on operational telemetry.

18 Research Contribution and Engineering Novelty

The primary contribution is not an individual algorithm. It is the integration of several complementary forms of evidence under a safety-oriented control architecture:

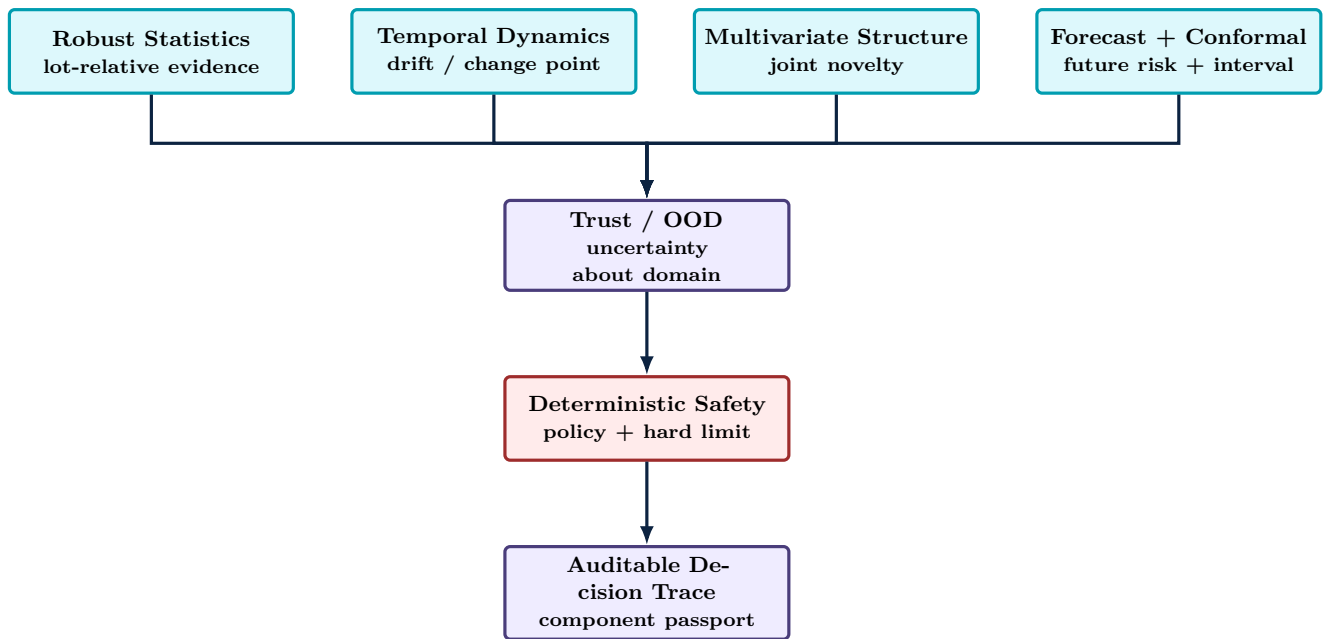


Figure 4: Research-to-engineering synthesis: heterogeneous statistical evidence is useful only after trust gating and deterministic safety policy.

The resulting system addresses four failure modes that a simple classifier can conflate:

1. an unusual part in an otherwise normal lot;
2. a slowly degrading part that has not yet exceeded a static limit;
3. a part whose multivariate trajectory is abnormal even when individual channels look acceptable;
4. a part presented under an unfamiliar domain where the model should reduce confidence rather than hallucinate certainty.

The contribution is therefore best described as **an interpretable, lot-aware, time-aware, uncertainty-aware screening architecture with an explicit safety boundary**.

19 Review Checklist

A reviewer can validate the project in the following order:

1. **Start with Figure 1** to understand the separation between AI evidence and deterministic safety.
2. **Inspect the generator contract** in Section 4 and verify that labels are not model inputs.
3. **Inspect the benchmark protocol** in Section 12: lot-grouped split, validation-only calibration, eight origins, three seeds and locked test.
4. **Read the benchmark results** in Section 13 and distinguish empirical synthetic metrics from future operational validation.
5. **Open a component passport in the frontend** and follow the nine-stage evidence trace from telemetry to final disposition.
6. **Inspect the hard override** and confirm that an authoritative physical-limit violation cannot be overruled by a model prediction.
7. **Inspect the artifact manifests and configuration hashes** before accepting a displayed result as part of the reproducibility chain.

20 Final System Definition

ACS =

Ingestion + Canonicalization + Robust Features + Module A Anomaly + Module B Forecast +
OOD / Trust + Safety Policy + Hard Override + Explainability + Audit

Authoritative physical-limit violation $\Rightarrow$ REJECT.

The final benchmark establishes a clear internal result: the complete pipeline, rather than a single isolated model, successfully detects the injected defect population without measured critical escapes under the locked synthetic test protocol while keeping the healthy false-positive rate at 1.52%. The forecast branch provides uncertainty-aware endpoint evidence, the trust layer prevents unfamiliar data from being silently interpreted as a defect, and the frontend exposes the entire decision chain rather than only a final label.

The remaining validation step is straightforward and intentionally external to this document: run the same architecture on approved operational telemetry, with authoritative limits and a truly held-out hardware population. Only after that step can real-hardware DPPM, chamber-hour savings, qualification claims, and field-reliability metrics be reported.

A Reference Equations

$$\text{Robust } z: \quad z = \frac{x - \text{median}(x)}{1.4826 \text{ MAD}(x) + \epsilon}$$

$$\text{Slope:} \quad s_k = \frac{x_k - x_{k-1}}{t_k - t_{k-1}}$$

$$\text{Acceleration:} \quad a_k = \frac{s_k - s_{k-1}}{t_k - t_{k-1}}$$

$$\text{Mahalanobis:} \quad D_M(x) = \sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)}$$

$$\text{Logistic stacker:} \quad p_A = \sigma(\beta_0 + \beta^\top \mathbf{s})$$

$$\text{Forecast interval:} \quad I_{168} = [\hat{x}_{168} - q_{1-\alpha}, \hat{x}_{168} + q_{1-\alpha}]$$

$$\text{Upper-limit crossing:} \quad \hat{x}_{168} + q_{1-\alpha} > L_{max}$$

$$\text{Hard safety invariant:} \quad x > L_{max} \vee x < L_{min} \Rightarrow \text{REJECT.}$$

B Final Benchmark Configuration Summary

```
Generator
parts = 400
lots = 80
parts_per_lot = 5
seed = 26170
defect_base_rate = 0.25
families = 6
lot_grouped_split = true
correlation_target_quiescent_input_leakage = 0.4
missingness_random_rate = 0.05
missingness_block_part_rate = 0.10
```

```

accelerating_coefficient = 0.002

Benchmark
seeds = [20260831, 20260832, 20260833]
origins_h = [12,24,48,72,96,120,144,168]
bootstrap = 1000
test_set_locked = true
threshold_tuning_on_test = false

Headline empirical results
defect_recall = 1.0000
false_negative_rate = 0.0000
healthy_false_positive_rate = 0.0151515
precision = 0.9333333
pr_auc = 0.7527006
critical_escapes = 0
conformal_coverage = 1.0000

```

The configuration block above is a compact documentation summary. The submitted repository/configuration files remain the authoritative machine-readable source.

C Note from Arch Rival

60-second evidence-first narration

“ACS does not replace engineering limits with AI. It uses AI to generate additional evidence before the limit is reached. We compare every part against its lot, measure temporal degradation, capture multivariate interactions, forecast the 168-hour state with conformal uncertainty, and explicitly measure domain novelty. A deterministic safety policy then decides whether the evidence supports SAFE, REVIEW, REJECT or UNKNOWN. The physical-limit override is independent of the model. In the locked synthetic benchmark, the final system achieved 100 percent defect recall, zero critical escapes, a 1.52 percent healthy false-positive rate, 93.33 percent precision and 0.753 PR-AUC, with 100 percent empirical conformal coverage across the reported evaluation. The next validation step is operational telemetry—not a hidden claim that the synthetic benchmark is already flight qualification.”

D Submission Integrity Note

The visual frontend may contain multiple presentation surfaces (overview, matrix, passport, evidence, audit). This dossier deliberately treats the final benchmark artifact as the authoritative quantitative source and avoids copying legacy/demo-only values into the results tables. This prevents a polished UI from becoming a source of metric drift.

End of ACS SIH 2026 Technical Dossier

All benchmark numbers are explicitly scoped to the controlled evaluation described above.